exp1 - HTML & CSS (map, frames, inline, internal, external)

```
<!DOCTYPE html>
<html>
<head>
<title>College Event Registration</title>

<link rel="stylesheet" href="style.css">

<style>
body {
   background-color: #f4f8ff;
   font-family: Arial, sans-serif;
}

h1 {
   text-align: center;
   color: navy;
}

.nav {
   text-align: center;
}

.nav a {
   margin: 0 10px;
   text-decoration: none;
   font-weight: bold;
}
```

```
.form-box {
    text-align: center;
}

.map-box {
    text-align: center;
    margin-top: 20px;
}

.map-box img {
    border: 2px solid #333;
    max-width: 100%;
    height: auto;
}
</style>
</head>

<body>

<h1> 🎓 College Event Registration Portal</h1>

<p style="text-align:center; color: green; font-weight: bold;">
Register for upcoming college events and explore campus locations.
</p>

<hr>
```

```
<div class="nav">
<a href="#register">Register</a> |
<a href="#map">Campus Map</a> |
<a href="https://www.google.com" target="_blank">External Link</a>
</div>

<hr>

<h2 id="register">📋 Event Registration Form</h2>

<form class="form-box" action="#" method="post">
Name: <input type="text" name="name" required><br><br>
Email: <input type="email" name="email" required><br><br>

Event:
<select name="event" required>
<option value="">Select Event</option>
<option>Tech Fest</option>
<option>Cultural Fest</option>
<option>Sports Meet</option>
</select>
<br><br>

<button type="submit">Register</button>
</form>

<hr>
```

```
<h2 id="map">🗺 Campus Map with Hotspots</h2>

<div class="map-box">
<img src="campus-map.jpg" usemap="#campusmap" alt="Campus Map">

<map name="campusmap">
<area shape="rect" coords="40,40,180,150"
href="https://en.wikipedia.org/wiki/Library"
target="_blank"
alt="Library"
title="Library">

<area shape="rect" coords="220,60,380,170"
href="https://en.wikipedia.org/wiki/Auditorium"
target="_blank"
alt="Auditorium"
title="Auditorium">

<area shape="rect" coords="120,200,280,330"
href="https://en.wikipedia.org/wiki/Cafeteria"
target="_blank"
alt="Cafeteria"
title="Cafeteria">
</map>
</div>

<hr>
```

<h2> 📺 Event Preview Frame</h2>

<iframe

width="560"

height="315"

src="https://www.youtube.com/embed/ysz5S6PUM-U"

title="YouTube video"

allowfullscreen>

</iframe>

<hr>

<footer>

© 2026 College Event Portal

</footer>

</body>

</html>

exp2 - JavaScript validation

```
<!DOCTYPE html>
<html>
<head>
<title>Elegant Registration Form</title>

<style>
body {
    margin: 0;
    padding: 0;
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background: linear-gradient(to right, #eef2f3, #dfe9f3);
}

h1 {
    text-align: center;
    margin-top: 40px;
    color: #2c3e50;
    font-weight: 600;
}

.form-box {
    width: 420px;
    margin: 40px auto;
    padding: 30px;
    background: #ffffff;
    border-radius: 12px;
    box-shadow: 0 10px 25px rgba(0, 0, 0, 0.08);
```

```
}

label {
    display: block;
    margin-top: 15px;
    font-weight: 500;
    color: #34495e;
}

input, textarea {
    width: 100%;
    padding: 10px;
    margin-top: 6px;
    border-radius: 6px;
    border: 1px solid #ccd6dd;
    font-size: 14px;
}

button {
    width: 100%;
    margin-top: 25px;
    padding: 12px;
    background-color: #2c3e50;
    color: white;
    border: none;
    border-radius: 8px;
    font-size: 16px;
    cursor: pointer;
```

```
}
</style>

<script>
function validateForm() {

    var fname = document.getElementById("fname").value.trim();
    var lname = document.getElementById("lname").value.trim();
    var password = document.getElementById("password").value;
    var email = document.getElementById("email").value.trim();
    var mobile = document.getElementById("mobile").value.trim();
    var address = document.getElementById("address").value.trim();

    var namePattern = /^[A-Za-z]+$/;
    var emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    var mobilePattern = /^[0-9]{10}$/;

    if (!namePattern.test(fname) || fname.length < 6) {
        alert("First Name must contain only alphabets and be at least 6 characters.");
        return false;
    }

    if (password.length < 6) {
        alert("Password must be at least 6 characters long.");
        return false;
    }

    if (!emailPattern.test(email)) {
```

```
        alert("Enter valid email (name@domain.com).");
        return false;
    }

    if (!mobilePattern.test(mobile)) {
        alert("Mobile must be exactly 10 digits.");
        return false;
    }

    if (lname === "") {
        alert("Last Name cannot be empty.");
        return false;
    }

    if (address === "") {
        alert("Address cannot be empty.");
        return false;
    }

    alert("Form submitted successfully!");
    return true;
}
</script>
</head>

<body>

<h1>Registration Form</h1>
```

```
<div class="form-box">
<form onsubmit="return validateForm()">

<label>First Name</label>
<input type="text" id="fname" required>

<label>Last Name</label>
<input type="text" id="lname" required>

<label>Password</label>
<input type="password" id="password" required>

<label>E-mail</label>
<input type="email" id="email" required>

<label>Mobile Number</label>
<input type="tel" id="mobile" required>

<label>Address</label>
<textarea id="address" rows="3" required></textarea>

<button type="submit">Submit</button>
</form>
</div>
</body>
</html>
```

exp3 - Hello World Servlet

```
import java.io.*;

import javax.servlet.*;

import javax.servlet.http.*;


public class HelloServlet extends HttpServlet{

protected void doGet(HttpServletRequest req,HttpServletResponse res)throws
ServletException,IOException{

res.setContentType("text/html");

PrintWriter out=res.getWriter();

out.println("Hello, World!");

}

}
```

exp4 - Form + Servlet

index.html

```
<!DOCTYPE html>
<html>
<body>
<form action="display" method="post">
<input name="name" placeholder="Name"><br>
<input name="age" placeholder="Age"><br>
<input type="submit">
</form>
</body>
</html>
```

DisplayServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class DisplayServlet extends HttpServlet{
protected void doPost(HttpServletRequest req,HttpServletResponse res)throws
ServletException,IOException{
res.setContentType("text/html");
PrintWriter out=res.getWriter();
out.println("Name: "+req.getParameter("name"));
out.println("Age: "+req.getParameter("age"));
```

}}

exp5 - GET vs POST

index.html

```
<!DOCTYPE html>
<html>
<body>
<form action="method" method="get">
<input name="data">
<input type="submit" value="GET">
</form>
<form action="method" method="post">
<input name="data">
<input type="submit" value="POST">
</form>
</body>
</html>
```

MethodServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class MethodServlet extends HttpServlet{
protected void doGet(HttpServletRequest req,HttpServletResponse res)throws
ServletException,IOException{
```

```
res.getWriter().println("GET: "+req.getParameter("data"));

}

protected void doPost(HttpServletRequest req,HttpServletResponse res)throws
ServletException,IOException{

res.getWriter().println("POST: "+req.getParameter("data"));

}

}
```

exp6 - Session tracking

```
import java.io.*;

import javax.servlet.*;

import javax.servlet.http.*;


public class SessionServlet extends HttpServlet{

protected void doPost(HttpServletRequest req,HttpServletResponse res)throws
ServletException,IOException{

HttpSession session=req.getSession();

session.setAttribute("user",req.getParameter("user"));

res.getWriter().println("Session Created for "+session.getAttribute("user"));

}

}
```

exp7 - Android Calculator

MainActivity.java

```
package com.example.calc;


import android.os.Bundle;

import android.view.View;

import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity{

EditText n1,n2;

TextView res;

protected void onCreate(Bundle b){

super.onCreate(b);

setContentView(R.layout.activity_main);

n1=findViewById(R.id.n1);

n2=findViewById(R.id.n2);

res=findViewById(R.id.res);

}

public void add(View
v){res.setText(""+(Integer.parseInt(n1.getText().toString())+Integer.parseInt(n2.getText().toStr
ing())));}

public void sub(View v){res.setText(""+(Integer.parseInt(n1.getText().toString())-
Integer.parseInt(n2.getText().toString())));}

public void mul(View
v){res.setText(""+(Integer.parseInt(n1.getText().toString())*Integer.parseInt(n2.getText().toStr
ing())));}
```

```
public void div(View
v){res.setText(""+(Integer.parseInt(n1.getText().toString())/Integer.parseInt(n2.getText().toStri
ng())));}

}
```

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

android:orientation="vertical" android:layout_width="match_parent"
android:layout_height="match_parent">

<EditText android:id="@+id/n1"/>

<EditText android:id="@+id/n2"/>

<Button android:text="Add" android:onClick="add"/>

<Button android:text="Sub" android:onClick="sub"/>

<Button android:text="Mul" android:onClick="mul"/>

<Button android:text="Div" android:onClick="div"/>

<TextView android:id="@+id/res"/>

</LinearLayout>
```

exp8 - Change font color + toast

MainActivity.java

```
package com.example.app;

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity{
TextView t;
protected void onCreate(Bundle b){
super.onCreate(b);
setContentView(R.layout.activity_main);
t=findViewById(R.id.text);
}
public void change(View v){
t.setTextColor(Color.RED);
t.setTextSize(30);
Toast.makeText(this,"Changed",Toast.LENGTH_SHORT).show();
}
}
```

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

android:orientation="vertical" android:layout_width="match_parent"
android:layout_height="match_parent">

<TextView android:id="@+id/text" android:text="Hello"/>

<Button android:text="Change" android:onClick="change"/>

</LinearLayout>
```

exp9 - Send email

```
package com.example.mail;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity{
protected void onCreate(Bundle b){
super.onCreate(b);
setContentView(R.layout.activity_main);
}
public void send(View v){
Intent i=new Intent(Intent.ACTION_SEND);
i.setType("message/rfc822");
i.putExtra(Intent.EXTRA_EMAIL,new String[]{"test@example.com"});
i.putExtra(Intent.EXTRA_SUBJECT,"Subject");
i.putExtra(Intent.EXTRA_TEXT,"Body");
startActivity(Intent.createChooser(i,"Send Mail"));
}
}
```

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent" android:layout_height="match_parent">
<Button android:text="Send Email" android:onClick="send"/>
</LinearLayout>
```